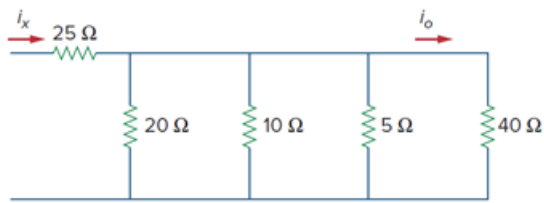


Problem

For the circuit in Fig. 2.90, $i_o = 3$ A. Calculate i_x and the total power absorbed by the entire circuit.

Figure 2.90:



Step-by-step solution

Step 1 of 3

Refer to the circuit diagram in Figure 2.90 in the textbook.

Calculate the voltage across the $40\ \Omega$ resistor.

$$\begin{aligned} V_{40\Omega} &= i_o (40\ \Omega) \\ &= (3\ \text{A})(40\ \Omega) \\ &= 120\ \text{V} \end{aligned}$$

Voltage across the parallel resistors is same.

Calculate the current through $5\ \Omega$ resistor.

$$\begin{aligned} i_{5\Omega} &= \frac{120\ \text{V}}{5\ \Omega} \\ &= 24\ \text{A} \end{aligned}$$

Step 2 of 3

Calculate the current through $10\ \Omega$ resistor.

$$\begin{aligned} i_{10\Omega} &= \frac{120\ \text{V}}{10\ \Omega} \\ &= 12\ \text{A} \end{aligned}$$

Calculate the current through $20\ \Omega$ resistor.

$$\begin{aligned} i_{20\Omega} &= \frac{120\ \text{V}}{20\ \Omega} \\ &= 6\ \text{A} \end{aligned}$$

Now, calculate the current i_x .

$$\begin{aligned} i_x &= i_{20\Omega} + i_{10\Omega} + i_{5\Omega} + i_o \\ &= 6\ \text{A} + 12\ \text{A} + 24\ \text{A} + 3\ \text{A} \\ &= 45\ \text{A} \end{aligned}$$

Therefore, the required current, i_x is 45 A.

Step 3 of 3

Calculate the power absorbed by the entire circuit.

$$\begin{aligned}\sum i^2 R &= \left[(45 \text{ A})^2 (25 \Omega) + (6 \text{ A})^2 (20 \Omega) + (12 \text{ A})^2 (10 \Omega) + \right. \\ &\quad \left. (24 \text{ A})^2 (5 \Omega) + (3 \text{ A})^2 (40 \Omega) \right] \\ &= 50625 \text{ W} + 720 \text{ W} + 1440 \text{ W} + 2880 \text{ W} + 360 \text{ W} \\ &= 56.03 \text{ kW}\end{aligned}$$

Therefore, the power absorbed by the entire circuit is 56.03 kW.